

Two Test Plans for Cure Time

AP Statistics · Unit 4 · Topic 4.9 · B: 2027-01-28 · E: 2027-03-17 · TEACHER KEY

CED TETHER

- **Skill 1.E** — Identify an appropriate inference method for significance tests.
- **Skill 1.F** — Identify null and alternative hypotheses.
- **Skill 4.C** — Verify that inference procedures apply in a given situation.
- **EU VAR-7** — The t-distribution may be used to model variation.
- **LO VAR-7.F** — Identify an appropriate selection of a testing method for a difference of two population means. [Skill 1.E]
- **EK VAR-7.F.1** — For a quantitative variable, the appropriate test for a difference of two population means is a two-sample t-test for a difference of two population means.
- **LO VAR-7.G** — Identify the null and alternative hypotheses for a difference of two population means. [Skill 1.F]
- **EK VAR-7.G.1** — The null hypothesis for a two-sample t-test for a difference of two population means, μ_1 and μ_2 , is: $H_0: \mu_1 - \mu_2 = 0$, or $H_0: \mu_1 = \mu_2$. The alternative hypothesis is $H_a: \mu_1 - \mu_2 < 0$, or $H_a: \mu_1 - \mu_2 > 0$, or $H_a: \mu_1 - \mu_2 \neq 0$, or $H_a: \mu_1 > \mu_2$, or $H_a: \mu_1 < \mu_2$, or $H_a: \mu_1 \neq \mu_2$.
- **LO VAR-7.H** — Verify the conditions for the significance test for the difference of two population means. [Skill 4.C]
- **EK VAR-7.H.1** — In order to make statistical inferences when testing a difference between population means, we must check for independence and that the sampling distribution is approximately normal:
 - a. Individual observations should be independent: (i) Data should be collected using simple random samples or a randomized experiment; (ii) When sampling without replacement, check that $n_1 \leq 10\% N_1$ and $n_2 \leq 10\% N_2$.
 - b. **The sampling distribution of $\bar{x}_1 - \bar{x}_2$ should be approximately normal (shape): (i) If the observed distribution is skewed, both n_1 and n_2 should be greater than 30; (ii) If the sample size is less than 30, the distribution of the sample data should be free from strong skewness and outliers — checked for BOTH samples.**

Essential question: How do the original question and study design control a defensible two-mean test plan?

DOK-3 skill: 4.C · **FRQ pattern:** predata-tail-parameter-and-scope

DOK rationale: Part (c) coordinates the pre-data question, parameter order, sample structure, distribution evidence, and assignment limits to adjudicate two test plans and trace three misleading conclusions.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK: 1**
→ 3 **Minutes: 45**

Teacher does

- Ask when each plan fixed its direction and what design produced the data.
- Begin the 7.8 video at minute 5; return at minute 33 for parts (a)–(c).
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Compare the timing, notation, and design claims in the two plans.
- Complete the follow-along, finish (a)–(c), revise, and turn in.

Questions to ask

- What wording fixed the alternative before any sample was seen?
- Why inspect each sample separately when both sizes are 18?
- What does sampling support that assignment would have supported?

Adult role

Solo teacher. Trace the question and design to each inferential choice.

[WARN] WATCH FOR

- Choosing a lower tail only because the observed difference is negative.
- Writing hypotheses with sample means.
- Treating separate SRSs as random assignment of formula.

SCORING PART (c) — E / P / I

Expected elements

- **selects-predata-tail** (required) — Uses the nondirectional question to defend a two-sided alternative
- **uses-population-parameters** (required) — Uses population means and rejects fixed sample means in hypotheses
- **diagnoses-posthoc-tail** (required) — Explains how post-data tail selection makes evidence look stronger
- **uses-design-evidence** (required) — Separates valid sampling conditions from absent assignment
- **bounds-causal-claim** (required) — Limits scope and identifies line conditions as confounded

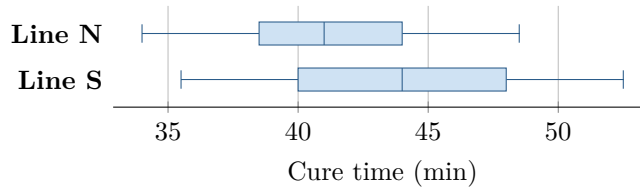
E	Defends the pre-data parameter test, diagnoses the post-data tail, uses design evidence, and bounds scope and causation.
P	States a valid test but incompletely connects timing, notation, conditions, scope, or causation.
I	Chooses a tail from the observed sign, tests sample means, or treats sampling as formula assignment.

[STAR] TODAY'S PROBLEM — annotated

Suppose existing lines N (new formula) and S (standard) make the same sealant. Independently, a team samples 18 of 600 recent N batches and 18 of 540 recent S batches. Cure-time summaries are $\bar{x}_N = 41.2$, $s_N = 3.6$, $\bar{x}_S = 44.0$, and $s_S = 4.1$; boxplots are shown. Before sampling it asked, “Do the two populations differ in mean cure time?” Formula was not assigned to lines.

Rhea: “Preserve the pre-data direction, use population means, and separate association from causation.”

Sol: “Since $41.2 < 44.0$, use a lower tail with sample means; significance proves formula caused faster curing.”



FIRST TAKE (5 min)

Which parts of each plan do the question’s timing and the design support? Name one detail to verify.

Key: Not graded. Require evidence about both timing and design.

Rules callout “LET THE QUESTION–NOT THE SAMPLE–SET THE TAIL (keep on the page)” is printed on the student sheet.

DOK 1 (a) Identify the response, both samples, and $\bar{x}_N - \bar{x}_S$. Did the pre-data question specify a direction?

Key: The response is batch cure time in minutes; the independent samples are 18 N and 18 S batches. The difference is $41.2 - 44.0 = -2.8$ minutes. The question specified no direction.

DOK 2 (b) Name the procedure; define both parameters; state H_0 in N-minus-S order; and verify independence, 10 percent, and shape.

Key: Use a two-sample t test. Let μ_N and μ_S be recent-population mean cure times; $H_0 : \mu_N - \mu_S = 0$. Independent SRSs and $18 \leq 60$, $18 \leq 54$ satisfy design checks. Since both $n = 18 < 30$, inspect both boxplots: each is roughly symmetric without outliers or strong skew.

DOK 3 (c) Adjudicate both plans. Defend the alternative and notation, diagnose selecting a tail from the observed order, and bound a significant result. Explain the misleading belief invited by each rejected choice.

Key: Use $H_a : \mu_N - \mu_S \neq 0$: the pre-data question asked about any difference. Unknown population means belong in hypotheses; fixed statistics do not. Selecting a lower tail after observing -2.8 favors the observed direction and overstates planned evidence. Significance could support a recent line-population mean difference, not formula causation, because formula and line are confounded. The rejected notation portrays statistics as unknown, the tail portrays direction as pre-specified, and causal wording portrays line differences as controlled.

EXIT

One thing the video changed about my first take: _____